

A SIMULATION-OPTIMIZATION APPROACH FOR ROBUST AIRCRAFT ROUTING

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ABSTRACT

To mitigate the impact of unpredictable disruptions, airlines are seeking to pro-actively design schedules that incorporate some robustness features. In this paper, we propose a novel simulation-optimization approach using a particle swarm optimization algorithm for solving the robust aircraft routing and flight re timing problem. The approach requires inserting buffer times between departure times of flights to improve the robustness of both aircraft and passengers connections. The proposed approach is based on discrete-event simulation for the evaluation of the solutions performance and a Particle Swarm Optimization (PSO) routine for guiding the search towards enhanced solutions. The results of computational experiments that were carried out on a real data are presented.

Keywords: Airline planning, Robustness, Delays, Aircraft routing, Particle Swarm Optimization (PSO), Simulation

1 INTRODUCTION:

The airline industry operates in an intensely competitive market where airline companies aim to provide best services at lower costs. Toward that aim, various approaches have been developed in the literature, making the application of optimization techniques one of the most impressive success stories of operations research. These approaches solve the operations' planning problems in a sequential or integrated procedures including mainly: schedule planning, fleet assignment, aircraft routing and crew scheduling.

Accordingly, these problems are usually modeled as large-scale (deterministic) mixed-integer programs and solved using optimization techniques. However, optimized schedules may be significantly affected by unpredictable disruptions. The sources of disruptions may include and not restrictively: aircraft shortages, crew shortages, passengers' shortages and inclement weather conditions. Therefore, airline delays are continuously increasing resulting in tremendous losses and have become an inevitable fact in airline transportation. Being aware of how huge is the profit running out from their hands, airlines are deeply investigating new approaches to mitigate these disruptions and enhance their profitability. Toward that aim, two broad strategies, with different purposes, have been developed: reactive strategies and proactive strategies .

The proactive strategies refer to a set of recovery measures, implemented after the occurrence of a disruption to repair airline schedules in a quick and cost effective manner. For the sake of details, we refer to Barnhart [1] who provides an overview of the most pertinent recovery strategies. However, proactive approaches are meant to be implanted in the planning stage in order to produce robust schedules that consider delays' impact and propose measures to prevent.

We focus throughout this paper on developing a proactive approach that generates robust plans less sensitive to delays caused by late-arriving flights. It is worth mentioning that delays caused by late arriving flights are the primary cause of delays. In fact, it totaled 42.1 % of total minute delays recorded in 2013 by US carriers. This paper includes the following contributions:

1. Given a flight schedule together with the associated fleet assignment and passengers itineraries, and given connection buffer time for each pair of flights, we develop a two-stage Mixed Integer Quadratic Programming (MIQP) model that aims at revising the flight departure times and assigning aircraft to flights while satisfying as much as possible the connection buffer times.
2. We propose a particle swarm optimization (PSO) algorithm that helps to determine the best connection buffer times to be inserted between consecutive flights.
3. We present the experimental results of a Monte-Carlo simulation study which demonstrate that the proposed schedules are less vulnerable to disruptions caused by late-arriving aircraft.

The remainder paper is organized as follow. In section 2, we propose a brief review of the literature dealing with robust airline operations planning models while in section 3, we describe the robust aircraft routing and retiming problem (RARR). Next, in section 4, we introduce the simulation-optimization approach used to determine the most effective connections' buffers times while in section 5, we present some numerical experiments. Finally, in section 6 we present some concluding remarks and directions for future research.

2 LITERATURE REVIEW

The concept of robustness matches with the ability to mitigate the impact of sudden disruptions when they occur. Incorporating robustness in airline schedule can be achieved by including some specific robustness-inducing features. We consider two aspects of robustness: stability and flexibility (Dück et al. [2]). Stability in airlines terminology refers to the insensitivity of the schedule to minor changes in the operating environment. Besides, the concept of flexibility refers to the ability of a schedule to adapt to external changes, while maintaining satisfactory performance.

2.1 Approaches used for generating stable plans

Stable airline schedules are attributed to likely remain feasible and near- cost optimal in case of disruption. A common practice to build stable airline schedules consists on adding buffer times to the most frequently delayed flights in order to prevent from a delay propagation in the network. In this regard, Ehrgott and Ryan [3] addressed the issue of constructing stable crew schedules for a domestic New Zealand airline. Stability was induced in the model through penalizing short buffers between consecutive legs in each crew pairing. Moreover, Lan et al. [4] proposed two distinct models for building stable aircraft routes through an optimal slack allocation. In the first model, they focused on minimizing the total expected delay of the developed aircraft routes, while the second one aims to to minimize the expected number of disrupted passengers.

Furthermore, Aloulou et al. [5] proposed a mixed integer programming (MIP) model to improve robustness of aircraft schedules through flight re-timing and subject to a fixed fleet and passengers' itineraries. Alternatively, Cadarsoa and Marín [6] developed a multi-objective integrated robust approach for airline scheduling, while considering flight frequencies and timetable. The timetable is designed in such way to insert enough slack times to absorb delays and hence reducing expected mis-connected passengers.

2.2 Approaches used for generating flexible plans

Embedding flexibility features in airline plans consists for the most part on maximizing the number of opportunities to easily recover from a disruption. We introduce in the following the most pertinent approaches to ensure flexibility in the airline planning stage:

1. Short cycles and hub connectivity (Rosenberger et al. [7]): In case of disruptions, airlines opt for some recovery measure to preserve the feasibility of their schedules, including but not restrictively canceling flights. Canceling a flight may be a painful solution since it requires the cancellation of the downstream leg on the same aircraft route but it spares airline the huge recovery costs including ferrying and accommodating passengers' costs. Hence, crew pairings and aircraft rotations with many short cycles are more insensitive to flight cancellations.
2. Station purity (Smith and Johnson [8]): The station purity technique consists on limiting the number of fleet types or crew compatible families operating in each spoke station. The motivation behind this technique is to create more swap opportunities of aircraft or crews, to cover operational disruptions. The limit on the number of fleets, respectively crew families, serving each station is called station's purity level.
3. Aircraft swap opportunities (Burke et al. [9]): This strategy seeks to maximize the number of swap opportunities to generate robust aircraft assignment.

3 A TWP-STAGE MIQP APPROACH FOR ROBUST AIRCRAFT ROUTING

In this section, we present a robust aircraft routing and re-timing (RARR) problem using a two-stage mixed integer quadratic programming network-flow model. Given a flight schedule together with the associated fleet assignment and passengers' itineraries, and given a connection buffer time for each pair of flights, the model determines both aircraft routes and flights' departure times while preventing the allocation of less time than the given connection buffer time for each two consecutive flights. The amount of buffer times for aircraft and passengers connections is first set arbitrarily, and is then adjusted iteratively using a particle swarm optimization (PSO) algorithm (see §4).

3.1 Notation

Let F be a set of m available aircraft with φ different aircraft types, in which all aircraft have the same seating configuration. Denote by $F_f \subset F$, the subset of aircraft that belong to aircraft type f and by θ_f their corresponding turn-time ($f = 1, \dots, \varphi$). Let L , indexed by j , denotes a set of n flights to be flown during the planning horizon (typically, one day). Each flight $j \in L$ is defined by its origin and destination stations, its duration d_j and a set of possible departure times I_j . If departure time $p \in I_j$ is selected, then the aircraft will take off at t_p^j and will land at $t_p^j + d_j$. Moreover, we define $L_f \subset L$ as the set of flights that are assigned to aircraft type f . Let S , indexed by s , be the set of stations. In order to take into account the aircraft origin location, define, for each station s and each aircraft type f , two sets of zero-duration dummy flights $L_f^{i(s)}$ and $L_f^{o(s)}$ that correspond to flight landing at and departing from station s , respectively. For each flight $j \in L_f^{i(s)}$ ($j \in L_f^{o(s)}$), we associate a single departure time that corresponds to the beginning (the end) of the planning horizon. Moreover, for each station s , we define A_f^s and B_f^s as the sets of flights that are assigned to aircraft of type f , and that arrive to and depart from station s , respectively. These sets include the dummy flights $L_f^{i(s)}$ and $L_f^{o(s)}$, respectively. For each pair of flights $(j, k) \in L_f \times L_f$, that are assigned to the same aircraft type f while having compatible stations (that is, the arrival station of flight j coincides with the departure station of flight k), we define I_{jk} as the set of compatible departure time pairs $(p, q), p \in I_j, q \in I_k$ (that is, those that allow an aircraft to serve flight k departing at t_q^k , immediately after flight j departing at t_p^j). For each pair of flights $(j, k) \in L_f \times L_f$, denote by Δ_{jk} the given buffer time to be introduced between flights j and k . Let γ_{pq}^{jk} be the penalty that maps the gap between the real buffer time, denoted by s_{jk} and the given one (Δ_{jk}). Moreover, for each aircraft connections, the real buffer time is given by:

$$s_{pq}^{jk} = t_q^k - (t_p^j + d_j + \theta_j) \quad \forall (j, k) \in L_f \times L_f, (p, q) \in I_{jk}$$

In order to respect as much as possible the given buffer time between flight j and k , the penalty

term γ_{pq}^{jk} is expressed as follows:

$$\gamma_{pq}^{jk} = \begin{cases} (\Delta_{jk} - s_{jk})^2 & \text{if } s_{jk} < \Delta_{jk} \\ 0 & \text{otherwise} \end{cases} \quad \forall (j, k) \in L_f \times L_f, (p, q) \in I_{jk}$$

Furthermore, let A the set of aircraft connecting flights, obtained after solving the routing problem and $\bar{\Delta}_{jk}$ their associated buffer times. To account for the flow of connecting passengers, we denote by T the set of connecting flights (in practice, T may be restricted to the set of passenger connections), and ϵ_{jk} the estimated minimum time for passengers connecting between flights j and k , for $(j, k) \in T$. Likely in aircraft connection problem, the real buffer time is given by:

$$\sigma_{jk} = t_k - (t_j + d_j + \epsilon_{jk}) \quad \forall (j, k) \in T$$

Where t_j and t_k represent the selected departure times of flights j and k , respectively.

3.2 Variables

x_j^p : binary decision variable that takes the value 1 if the departure time $p \in I_j$ is selected for flight $j \in L$, and 0 otherwise.

z_{pq}^{jk} : binary decision variable that takes the value 1 if the same aircraft is assigned to flight k departing at time q immediately after flight j departing at time p , and 0 otherwise.

y_{jk} : continuous decision variables that define the selected slack between passengers connecting flights j and k .

In the first stage of the proposed two-stage RARR problem, aircraft routes are built to incorporate buffer times between two consecutive aircraft connections. Given the aircraft routes and the selected aircraft buffers from the first stage problem, the second stage problem determines the departure time of each scheduled flight while judiciously allocating buffers for passengers' connections. The two-stage RARR problem can be stated as follows:

3.3 Model 1.1. Two-stage RARR problem considering aircraft connections

$$\text{Minimize : } \sum_{f=1}^{\varphi} \sum_{s \in S} \sum_{j \in A_f^s} \sum_{k \in B_f^s} \sum_{(p,q) \in I_{jk}} \gamma_{pq}^{jk} z_{pq}^{jk} \quad (1)$$

subject to :

$$\sum_{q \in I_k} t_q^k x_q^k - \sum_{p \in I_j} t_p^j x_p^j \geq d_j + \epsilon_{jk}, \quad (j, k) \in T \quad (2)$$

$$\sum_{p \in I_j} x_p^j = 1, \quad j \in L \quad (3)$$

$$\sum_{j \in A_f^s} \sum_{(p,q) \in I_{jk}} z_{pq}^{jk} = x_q^k, \quad f = 1, \dots, \varphi, \quad s \in S, \quad k \in B_f^s, \quad q \in I_k \quad (4)$$

$$\sum_{k \in B_f^s} \sum_{(p,q) \in I_{jk}} z_{pq}^{jk} = x_p^j, \quad f = 1, \dots, \varphi, \quad s \in S, \quad j \in A_f^s, \quad p \in I_j \quad (5)$$

$$x_p^j \in \{0, 1\}, \quad j \in L, \quad p \in I_j \quad (6)$$

$$z_{pq}^{jk} \in \{0, 1\}, \quad f = 1, \dots, \varphi, \quad j \in L_f \cup_{s \in S} L_f^{i(s)}, \quad k \in L_f \cup_{s \in S} L_f^{o(s)}, \quad (p, q) \in I_{jk} \quad (7)$$

The objective function (1) is to minimize penalties induced by buffers' violation for each aircraft connection. Constraint (2) enforces that after flight re-timing, the passengers can still make their planned connections within the required connection time ϵ_{jk} . Constraint (3) ensures that an unique departure time is assigned to each flight, while constraints (4) and (5) guarantee that each flight has exactly one predecessor and one successor.

3.4 Model 1.2. Two-stage RARR problem considering passengers' connections

$$\text{Minimize : } \sum_{(j,k) \in T} y_{jk}^2 \quad (8)$$

subject to :

$$y_{jk} + \sum_{q \in I_k} t_q^k x_q^k - \sum_{p \in I_j} t_p^j x_p^j \geq \Delta_{jk} + d_{j+} \epsilon_{jk}, \quad (j, k) \in T \quad (9)$$

$$\sum_{p \in I_j} x_p^j = 1, \quad j \in L \quad (10)$$

$$\sum_{q \in I_k} t_q^k x_q^k - \sum_{p \in I_j} t_p^j x_p^j \geq \bar{\Delta}_{jk} + d_{j+} \theta_j, \quad (j, k) \in A \quad (11)$$

$$x_p^j \in \{0, 1\}, \quad j \in L, \quad p \in I_j \quad (12)$$

$$y_{jk} \geq 0 \quad \forall (j, k) \in T \quad (13)$$

The objective function (8) is to minimize penalties induced by buffers' violation for each passengers' connection. Constraint (9) defines buffer time to be allocated between each connection. In particular, constraint (9) ensures that, after flight re-timing, passengers can still make their planned connections within the required connection time ϵ_{jk} . Constraint (10) ensures that a unique departure time is assigned to each flight, while constraint (11) is to maintain the feasibility of aircraft connections within the buffers inserted in the first stage.

4 THE SIMULATION-OPTIMIZATION APPROACH

The proposed simulation-optimization approach aims at building robust aircraft routes. It includes three main procedures: a particle swarm optimization (PSO) algorithm, a two-stage MIQP approach and a Monte-Carlo simulation study.

4.1 The PSO algorithm application:

In this study, we resort to a particle swarm optimization technique to find out a best set of buffer times, by repetitively improving an initial solution with regard to a given measure of quality. It is a population based metaheuristic with the ability to search large spaces of candidate solutions (Kennedy [10]).

The system is initialized with a population of candidate solutions (called also particles), each one corresponds to a set of buffer times generated randomly in the interval of $[\Delta_{\min}, \Delta_{\max}]$, where $\Delta_{\min}$ and $\Delta_{\max}$ correspond to the minimal and the maximal buffer times respectively. The considered interval is called a confinement interval. Each particle records its current position in the search space (fitness) as well as its best fitness achieved so far. We note by $pbest$ the best fitness achieved while $gbest$ refers to the best fitness, obtained so far among its neighbors.

The fitness is then improved iteratively by moving around particles according to a few simple formulas (Kennedy [10]). The process is repeated until a satisfactory solution is obtained.

4.2 A heuristic algorithm for solving the MIQP problem

At this step, the two-stage MIQP is solved to build aircraft routes and fix the flight departure time such that the given connection buffer times are respected as much as possible. A preliminary solution of the problem, developed above in the paper, has required a long computation time (around 180s) which may increase considerably the CPU time of the entire approach. A practical way to handle this issue is to look for a heuristic algorithm that may reduce the CPU time without altering the model tractability. Toward that aim, we propose an algorithm to be introduced in the first stage of the model, based

Algorithm 1 A linear relaxation algorithm for aircraft routing and re-timing problem.

- 1: Relax model (1) – (8) by setting the x-variables as continuous.
 - 2: Solve the relaxed model as a linear program.
 - 3: Generate new departure times for each flight leg, referred by t_j and obtained by $t_j = \sum_{p \in I_j} p x_p^j$.
 - 4: Build a reduced set of re-timing opportunities for each flight. The specified set includes only two values that correspond the upper and lower values of t_j in the original re-timing opportunities set.
 - 5: Solve the model (1) – (8) with the new re-timing opportunities.
 - 6: Solve the model (9) – (15) given the aircraft routes obtained from the first stage.
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on a linear programming relaxation technique, to effectively solve the MIQP problem. The specific algorithm is described as follows:

The proposed model has shown a tangible proof of efficiency, indeed, it succeeded to reduce the computation time by a ratio of 5 while altering the solution by only 0.5%.

4.3 Monte Carlo Simulation study

To evaluate the robustness of the candidate solutions derived by the PSO algorithm, we carried out a Monte Carlo simulation study, following the approach proposed by Aloulou et al. [5].

To assess the performance of the proposed solution, we consider the following performance measures:

- PM1: Total reactionary delay, when the delay is more than 15 min.
- PM2: Number of flights delayed by more than 15 min.
- PM3: Number of passengers delayed by more than 15 min.
- PM4: Number of passengers missing their connections.

5 COMPUTATIONAL RESULTS

To assess the tractability and the relevance of our simulation-optimization approach, we conducted a computational study using instances referred to by A and B, respectively. These instances were proposed by Amadeus, SAS for the ROADEF Challenge 2009. They represent one-day schedules of two existent airline companies. Table 1. displays the characteristics of the considered instances.

Table 1. Instances characteristics

Instance	A	B
Number of flights	464	1278
Number of aircraft	81	251
Number aircraft types	14	29
Number of stations	35	44
Number of passenger connections	642	1449

Table 2. Average performance measures.

Scenario	PM1(min)	PM2	PM3	PM4
A_0	99645.73	286.9	21708.79	1145,54
A_1	790,48	27,91	2335,64	47,37
A_2	545,8	20,19	1751,56	29,75
A_3	351,36	13,45	1190,32	15,56
A_4	386,06	14,73	1297,16	19,69
A_5	323,47	12,37	1086,07	17,44
A_6	299,34	11,69	1030,45	16,12
B_0	268612,57	772,18	59979,08	4490,57
B_1	1795,55	64,43	5195,68	179,25
B_2	1165,85	43,78	3584,87	106,68
B_3	761,77	29,73	2443,74	78,11
B_4	755,19	29,34	2434,58	75,78
B_5	535,04	21,15	1760	47,52
B_6	512,81	20,55	1726,72	46,43

Starting from instance A, respectively instance B, we built 6 instances $A_1, \dots, A_6$, that are obtained by defining for each leg $j \in L$ a time-window $[a_j - \delta, a_j + \delta]$ where a_j represents the genuine departure

time and parameter $\delta \in \{0, 5, 10, 15\}$ represents half of the time window width. Note that instance A_0 refers to the initial non-robust aircraft schedule. In instance A_1 , the flights' departure times are same as those of instance A_0 but aircraft routes are subject to modification and new routes will be built by our model. Next, we define for each instance a number $A_i (i = 2..6)$ a number v_i of alternative departure times (within the time-window) that are evenly spaced by r minutes (with $r \in \{0, 5, 10, 15\}$). Similarly, we proceed to build six instances $B_1, ..., B_6$ that are derived from the genuine instance B. With regard to the PSO algorithm, we set the size of our swarm (candidate particles) to 10. Each particle corresponds to the connection buffer times Δ_{jk} , to be assigned to each consecutive flights j and k . The initial particles are generated randomly among values in $[0, 80]$ while taking into account the position of the connection in the time horizon. Higher buffer times are given to early connections since they are more critical. Indeed, an initial delay on an early flight can propagate to the connecting flights, and spread the network causing more severe delays. Besides, we consider the fitness function as the total reactionary delay (PM1) obtained through the simulation study. Finally, we set a CPU time limit during which the algorithm will be run. It corresponds to 5000s for instance A, and to 10000s for instance B.

All our experiments were run on an i7 dual core 1.8 GHz Personal Computer with 8 GB RAM. In addition, the two-stage MIQP was solved to optimality, at each iteration and for each candidate solution, using CPLEX 12.6 solver with the default settings. Table 2 provides a summary of the main results. It presents the average performance measures PM1, PM2, PM3, and PM4 for non-robust and robust schedules of instances A and B.

Looking at Table 2, a striking observation is that, all measures of performance have tremendously decreased when integrating robustness features. For instance, the number of delayed flights are divided by a factor of around 10 for instance A, and respectively 12 for instance B. This shows that the proposed simulation-optimization approach builds optimized schedules that exhibit good performance and consistently outperform the genuine ones. Indeed, by avoiding tight connections and long rotations, we ensure that an initial delay can not spread in the network causing more severe delays.

Besides, in Table 3 we compare our results with others provided by Aloulou et al. [5], we prove that this approach better improves PM1, PM2 and PM4, while proposing worst results for the PM3.

Table 3. Comparison of performance measures with regard to Aloulou et al. (2013).

Sc	Aloulou et al. (2013)				Improvement(%)			
	PM1	PM2	PM3	PM4	PM1	PM2	PM3	PM4
A_1	800,42	28,22	2337,46	43,8	1,26	1,11	0,08	-7,54
A_2	591,81	21,75	1774,17	27,63	8,43	7,73	1,29	-7,13
A_3	384,96	14,34	1152,25	18,19	9,56	6,62	-3,20	16,90
A_4	379,14	14,25	1152,78	18,88	-1,79	-3,26	-11,1	-4,11
A_5	345,88	12,91	1057,68	16,08	6,93	4,37	-2,61	-7,80
A_6	324,48	12,27	999,1	17,43	8,40	4,96	-3,04	8,13
B_1	1812,76	64,91	5192,41	179,51	0,95	0,74	-0,06	0,14
B_2	1166,32	43,79	3438,91	108,38	0,04	0,02	-4,24	1,57
B_3	770,38	29,69	2257,83	80,35	1,12	-0,13	-8,23	2,79
B_4	760,66	29,4	2222,26	79,31	0,72	0,20	-9,55	4,45
B_5	538,37	21,13	1575,24	52,18	0,62	-0,09	-11,7	8,93
B_6	518,57	20,56	1508,26	51,84	1,11	0,05	-14,4	10,44

6 CONCLUSION

In this paper, we focused on building robust aircraft schedules. This problem is modeled as an objective programming problem, allowing the airline company to set different parameters as a measure of robustness to be optimized. Robustness is introduced into the schedule by assigning efficient buffer times

to aircraft and passengers connections while re-timing flight departures. We developed a simulation-optimization approach that combines a PSO algorithm, a two-stage MIQP solution and a Monte Carlo simulation. The obtained solution is a set of robust aircraft routes that minimize the reactionary delays. Computational results show the tractability and the efficiency of the proposed approach. They also provide empirical evidence that the optimized schedules are robust and thereby improving the on-time performance. Further improvement can be achieved by considering a multi-objective algorithm where a set of performance measures are optimized simultaneously.

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